

Linear Regression

Linear Regression is a **Supervised Machine Learning algorithm** used to predict **continuous numerical values** by modeling the relationship between independent variable(s) and dependent variable.

It assumes that the relationship between variables is **linear**.

General Idea:

We try to draw the best straight line that minimizes prediction error.

Used when:

- Output is continuous (price, salary, marks, temperature)
- We want interpretability
- We want to understand feature impact
- Data shows linear trend

Examples:

- House price prediction
- Salary prediction
- Sales forecasting
- Demand prediction

Types of Linear Regression

Simple Linear Regression

Used when there is **only one independent variable**

Model:

$$Y_p = mX + b$$

Where:

X = Independent variable

Y_p = Predicted value

m = Slope

b = Intercept

Graph: Straight line in 2D

Multiple Linear Regression

Used when there are **multiple independent variables**

Model:

$$Y_p = b_0 + b_1X_1 + b_2X_2 + b_3X_3 + \dots + b_nX_n$$

Example:

$$\text{Salary} = b_0 + b_1(\text{Experience}) + b_2(\text{Education}) + b_3(\text{Skills})$$

Here:

b_0 = Intercept

b_1, b_2, b_3 = Coefficients

Graph: Hyperplane in multi-dimensional space

Statistical Foundation

Mean of X:

$$\bar{X} = (\sum X) / n$$

Mean of Y:

$$\bar{Y} = (\sum Y) / n$$

Slope:

$$m = \sum (X - \bar{X})(Y - \bar{Y}) / \sum (X - \bar{X})^2$$

Intercept:

$$b = \bar{Y} - m\bar{X}$$

Intuition Behind Slope

Slope represents:

How much Y changes when X increases by 1 unit.

If $m = 2$

Then when X increases by 1 $\rightarrow$ Y increases by 2

Assumptions of Linear Regression

1. Linearity
Relationship between X and Y is linear
2. Independence
Observations are independent
3. Homoscedasticity
Constant variance of errors
4. No Multicollinearity (for multiple regression)
Independent variables should not be highly correlated
5. Normality of errors

Cost Function

Linear Regression uses:

Mean Squared Error (MSE)

$$MSE = (1/n) \sum (Y - Y_p)^2$$

Goal:

Minimize MSE

Why square?

- Removes negative sign
- Penalizes large errors more

Important Statistical Terms

Mean of X:

$$\bar{X} = (\sum X) / n$$

Mean of Y:

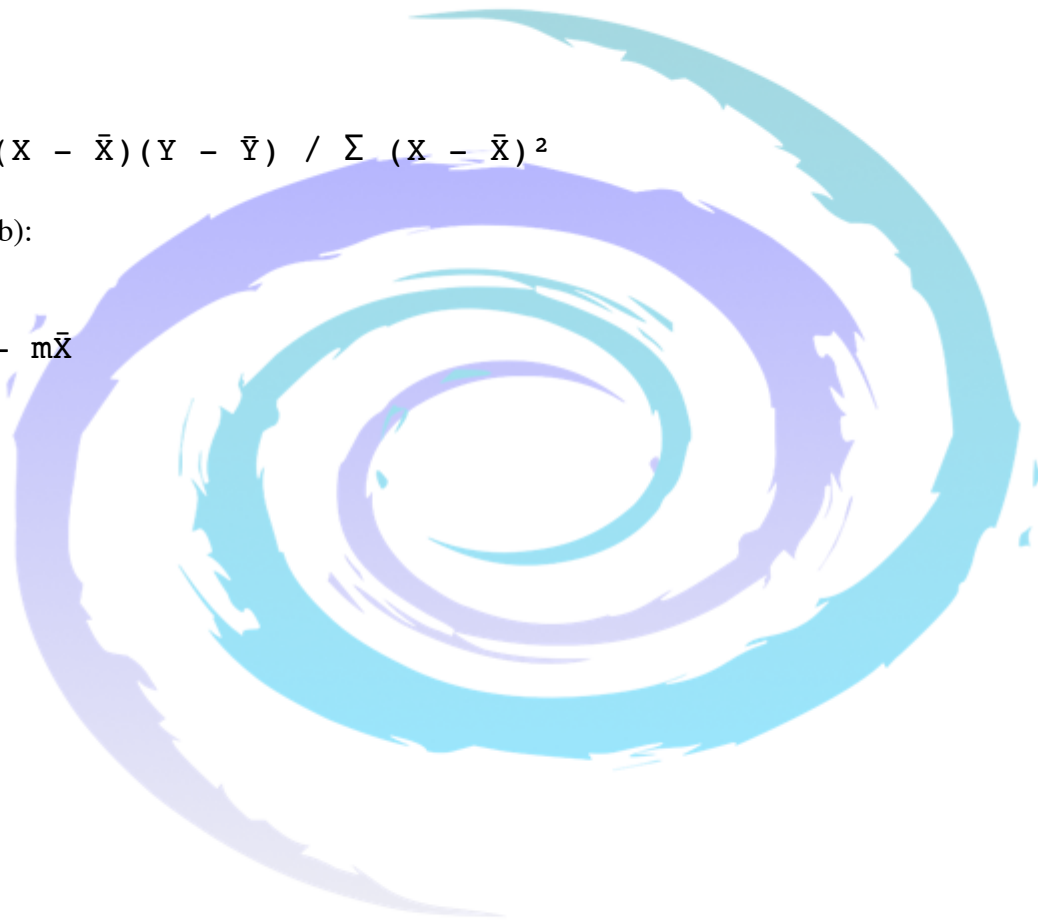
$$\bar{Y} = (\sum Y) / n$$

Slope (m):

$$m = \sum (X - \bar{X})(Y - \bar{Y}) / \sum (X - \bar{X})^2$$

Intercept (b):

$$b = \bar{Y} - m\bar{X}$$



Step-by-Step Calculation

Step 1: Calculate Means

$$X = [1, 2, 3, 4, 5]$$

$$Y = [3, 4, 2, 4, 5]$$

$$n = 5$$

$$\Sigma X = 15$$

$$\Sigma Y = 18$$

$$\bar{X} = 15 / 5 = 3$$

$$\bar{Y} = 18 / 5 = 3.6$$

Step 2: Create Calculation Table

X	Y	X - $\bar{X}$	Y - $\bar{Y}$	(X - $\bar{X}$)(Y - $\bar{Y}$)	(X - $\bar{X}$) ²
1	3	-2	-0.6	1.2	4
2	4	-1	0.4	-0.4	1
3	2	0	-1.6	0	0
4	4	1	0.4	0.4	1
5	5	2	1.4	2.8	4

Now sum:

$$\Sigma (X - \bar{X})(Y - \bar{Y}) = 1.2 - 0.4 + 0 + 0.4 + 2.8 = 4$$

$$\Sigma (X - \bar{X})^2 = 4 + 1 + 0 + 1 + 4 = 10$$

Step 3: Calculate Slope

$$m = 4 / 10$$

$$m = 0.4$$

Step 4: Calculate Intercept

$$b = \bar{Y} - m\bar{X}$$

$$b = 3.6 - (0.4 \times 3)$$

$$b = 3.6 - 1.2$$

$$b = 2.4$$

Final Regression Equation

$$Y_p = 0.4X + 2.4$$

Predicted Values (Y_p)

X	Actual Y	$Y_p = 0.4X + 2.4$
1	3	2.8
2	4	3.2
3	2	3.6
4	4	4.0
5	5	4.4

Residual (Error)

$$\text{Residual} = Y - Y_p$$

X	Y	Y_p	Residual
1	3	2.8	0.2
2	4	3.2	0.8
3	2	3.6	-1.6
4	4	4.0	0
5	5	4.4	0.6

Mean Squared Error (MSE)

Formula:

$$\text{MSE} = (1/n) \sum (Y - Y_p)^2$$

Squared residuals:

$$0.2^2 = 0.04$$

$$0.8^2 = 0.64$$

$$(-1.6)^2 = 2.56$$

$$0^2 = 0$$

$$0.6^2 = 0.36$$

$$\text{Sum} = 3.6$$

$$\text{MSE} = 3.6 / 5$$

$$\text{MSE} = 0.72$$

R-Squared (R²)

R² measures how much variation in Y is explained by X.

Formula:

$$R^2 = 1 - (\text{SS}_{\text{res}} / \text{SS}_{\text{tot}})$$

Where:

$$\text{SS}_{\text{res}} = \sum (Y - Y_p)^2$$

$$\text{SS}_{\text{tot}} = \sum (Y - \bar{Y})^2$$

Step 1: Calculate SS_tot

Y	Y - $\bar{Y}$	Square
3	-0.6	0.36
4	0.4	0.16
2	-1.6	2.56
4	0.4	0.16
5	1.4	1.96

$$\text{SS}_{\text{tot}} = 5.2$$

Step 2: SS_res

Already calculated = 3.6

Step 3: Calculate R^2

$$R^2 = 1 - (3.6 / 5.2)$$

$$R^2 = 1 - 0.6923$$

$$R^2 = 0.3077$$

Final $R^2 \approx 0.31$

Interpretation:

Model explains only 31% variance.

Weak linear relationship.

Conceptual Understanding

SS_tot → Total variation in Y

SS_res → Unexplained variation

$$SS_{exp} = SS_{tot} - SS_{res}$$

Explained Variation:

$$5.2 - 3.6 = 1.6$$

$$1.6 / 5.2 = 0.31$$

Same R^2 .

Training Process Internally

1. Initialize m and b randomly
2. Calculate Y_p
3. Compute error
4. Compute cost
5. Update m and b
6. Repeat

This is called Optimization.